



# HACKTHEBOX

## Detroit Become Human Writeup

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6<sup>th</sup> February 2025

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Difficulty: **Hard**

# Description

A DFIR analyst was notified of a possible incident on Forela's sysadmin machine. You are tasked to help the analyst in analysis to find the true source of this odd incident.

# Scenario

Alonzo Spire is fascinated by AI after noticing the recent uptick in usage of AI tools to help aid in daily tasks. He came across a sponsored post on social media about an AI tool by Google. The post had a massive reach, and the Page which posted had 200k + followers. Without any second thought, he downloaded the tool provided via the Post. But after installing it he could not find the tool on his system which raised his suspicions. A DFIR analyst was notified of a possible incident on Forela's sysadmin machine. You are tasked to help the analyst in analysis to find the true source of this odd incident.

# Artifacts Provided

Enter the artifacts provided along with their file hash here.

- detroitbecomehuman.zip -  
*aaf034c530c88a8c743364401fb8c518654f5d2c1660b8117302b70795e9101d*

# Skills Learnt

- NTFS Forensics
- Timeline creation
- File Recovery
- Disk Forensics
- Contextual Analysis
- Event Log Analysis

- Threat intelligence
- Browser forensics

## Initial Analysis

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To begin the analysis, the password-protected ZIP file was unlocked using the password `hacktheblue` .

Inside the archive we found what appears to be a `KAPE` extraction of a **Windows** C drive. These extractions typically contain numerous forensically relevant artifacts from a Windows system, such as registry hives, Windows event logs, prefetch files, browser history, \$MFT (Master File Table) file, user home directories, etc.

```
[achille@parrot]-[~/Triage/C]
└─ $ll
total 480M
drwx----- 1 achille achille  12 Mar 19  2024 '$Extend'
-rw-r--r-- 1 achille achille 480M Mar  8  2023 '$MFT'
drwx----- 1 achille achille 108 Mar 19  2024 '$Recycle.Bin'
drwx----- 1 achille achille 160 Mar 19  2024 Users
drwx----- 1 achille achille  62 Mar 19  2024 Windows
[achille@parrot]-[~/Triage/C]
└─ $tree -L 2
.
├── $Extend
│   ├── $J
│   └── $Max
├── $MFT
├── $Recycle.Bin
│   ├── S-1-5-18
│   └── S-1-5-21-3239415629-1862073780-2394361899-1104
├── Users
│   ├── Administrator
│   ├── alonzo.spire
│   ├── CyberJunkie
│   ├── Default
│   ├── defaultuser0
│   ├── happy.grunwald
│   └── Simon.stark
└── Windows
    ├── prefetch
    ├── ServiceProfiles
    └── System32

17 directories, 3 files
```

## Questions

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1. What is the full link of a social media post which is part of the malware campaign, and was unknowingly opened by Alonzo spire?

From the scenario, we know that Alonzo saw a social media post, downloaded a file from a link in that post, and installed it. So we focus our analysis first on artifacts related to web browsing, such as the browser history.

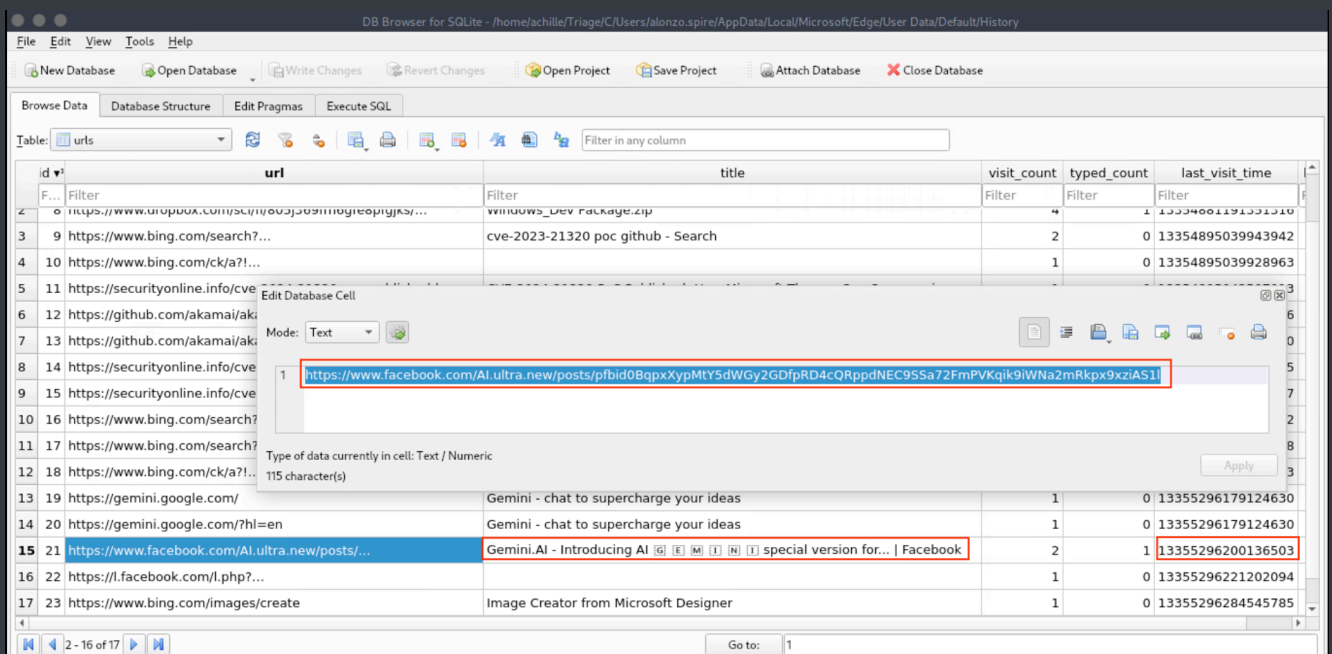
The majority of browsers store their history in a SQLite database and usually this file is called indeed History .

```
[achille@parrot]~[/Triage/C]
$ find . -type f -iname "history"
./Users/alonzo.spire/AppData/Local/Microsoft/Edge/User Data/Default/History
./Users/alonzo.spire/AppData/Local/Microsoft/Edge/User Data/Snapshots/113.0.1774.35/Default/History
./Users/alonzo.spire/AppData/Local/Microsoft/Edge/User Data/Snapshots/114.0.1823.43/Default/History
./Users/alonzo.spire/AppData/Local/Microsoft/Edge/User Data/Snapshots/116.0.1938.69/Default/History
[achille@parrot]~[/Triage/C]
$
```

There is a history file for the **Edge** browser, so we open it and search for anything related to social media posts.

Inside the `urls` table there are a few addresses, one of which appears to be a Facebook URL to a post about **Gemini AI**:

<https://www.facebook.com/AI.ultra.new/posts/pfbid0BqpxXypMtY5dWGy2GDfpRD4cQRppdNEC9SSa72FmPVKqik9iWNa2mRkpx9xziAS1l> .



The screenshot shows the DB Browser for SQLite interface. The 'urls' table is selected, and a modal window is open for editing a cell. The URL <https://www.facebook.com/AI.ultra.new/posts/pfbid0BqpxXypMtY5dWGy2GDfpRD4cQRppdNEC9SSa72FmPVKqik9iWNa2mRkpx9xziAS1l> is highlighted in the modal. The table data is as follows:

| id | url                                                   | title                                                                    | visit_count | typed_count | last_visit_time   |
|----|-------------------------------------------------------|--------------------------------------------------------------------------|-------------|-------------|-------------------|
| 2  | https://www.dropbox.com/sc/117603j303m10g9e0p1gks/... | windows_dev Package.zip                                                  | 4           | 1           | 13354601191351310 |
| 3  | https://www.bing.com/search?...                       | cve-2023-21320 poc github - Search                                       | 2           | 0           | 13354895039943942 |
| 4  | https://www.bing.com/ck/a?!...                        |                                                                          | 1           | 0           | 13354895039928963 |
| 5  | https://securityonline.info/cve/...                   |                                                                          |             |             |                   |
| 6  | https://github.com/akamai/akamai-edge-client          |                                                                          |             |             |                   |
| 7  | https://github.com/akamai/akamai-edge-client          |                                                                          |             |             |                   |
| 8  | https://securityonline.info/cve/...                   |                                                                          |             |             |                   |
| 9  | https://securityonline.info/cve/...                   |                                                                          |             |             |                   |
| 10 | https://www.bing.com/search?...                       |                                                                          |             |             |                   |
| 11 | https://www.bing.com/search?...                       |                                                                          |             |             |                   |
| 12 | https://www.bing.com/ck/a?!...                        |                                                                          |             |             |                   |
| 13 | https://gemini.google.com/                            | Gemini - chat to supercharge your ideas                                  | 1           | 0           | 13355296179124630 |
| 14 | https://gemini.google.com/?hl=en                      | Gemini - chat to supercharge your ideas                                  | 1           | 0           | 13355296179124630 |
| 15 | https://www.facebook.com/AI.ultra.new/posts/...       | Gemini AI - Introducing AI G E M I N I special version for...   Facebook | 2           | 1           | 13355296200136503 |
| 16 | https://l.facebook.com/l.php?...                      |                                                                          | 1           | 0           | 13355296221202094 |
| 17 | https://www.bing.com/images/create                    | Image Creator from Microsoft Designer                                    | 1           | 0           | 13355296284545785 |

Answer:

<https://www.facebook.com/AI.ultra.new/posts/pfbid0BqpxXypMtY5dWGy2GDfpRD4cQRppdNEC9SSa72FmPVKqik9iWNa2mRkpx9xziAS1l>

2. Can you confirm the timestamp in UTC when alonzo visited this post?

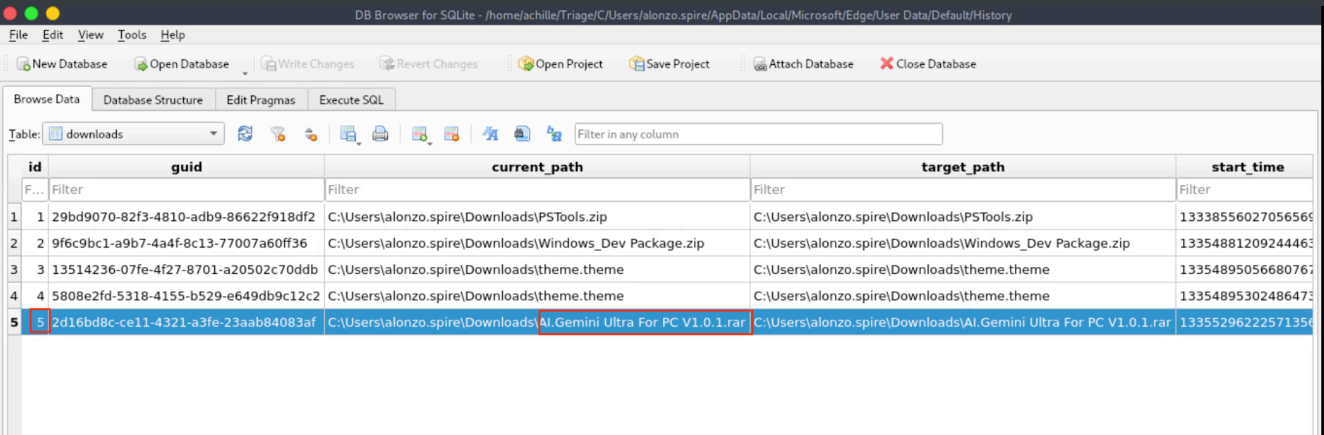
We can get the timestamp by looking at the `last_visit_time` column:  
13355296200136503 . This is a timestamp in **Chrome/WebKit** format, so we need to convert it to a UTC timestamp, and then we get 2024-03-19 04:30:00 .

```
[achille@parrot]--[~/Triage/C]
$ date -d @"((13355296200136503/1000000)-11644473600)" +"%Y-%m-%d %H:%M:%S"
2024-03-19 04:30:00
[achille@parrot]--[~/Triage/C]
$
```

Answer: 2024-03-19 04:30:00

3. Alonzo downloaded a file on the system thinking it was an AI Assistant tool. What is name of the archive file downloaded?

This information is also available in the browser history file, if we switch to the `downloads` table we can see 5 entries, the last one refers to the suspicious archive: AI.Gemini Ultra For PC V1.0.1.rar .



| id | guid                                 | current_path                                                      | target_path                                                       | start_time        |
|----|--------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|-------------------|
| 1  | 29bd9070-82f3-4810-adb9-86622f918df2 | C:\Users\alonzo.spire\Downloads\PSTools.zip                       | C:\Users\alonzo.spire\Downloads\PSTools.zip                       | 13338556027056566 |
| 2  | 9f6c9bc1-a9b7-4a4f-8c13-77007a60ff36 | C:\Users\alonzo.spire\Downloads\Windows_Dev Package.zip           | C:\Users\alonzo.spire\Downloads\Windows_Dev Package.zip           | 13354881209244463 |
| 3  | 13514236-07fe-4f27-8701-a20502c70ddb | C:\Users\alonzo.spire\Downloads\theme.theme                       | C:\Users\alonzo.spire\Downloads\theme.theme                       | 13354895056680767 |
| 4  | 5808e2fd-5318-4155-b529-e649db9c12c2 | C:\Users\alonzo.spire\Downloads\theme.theme                       | C:\Users\alonzo.spire\Downloads\theme.theme                       | 13354895302486473 |
| 5  | 2d16bd8c-ce11-4321-a3fe-23aab84083af | C:\Users\alonzo.spire\Downloads\AI.Gemini Ultra For PC V1.0.1.rar | C:\Users\alonzo.spire\Downloads\AI.Gemini Ultra For PC V1.0.1.rar | 13355296222571356 |

Answer: AI.Gemini Ultra For PC V1.0.1.rar

4. What was the full direct url from where the file was downloaded?

To answer this question, we need to note the download ID from the previous table ( 5 ) and then go to the `download_url_chains` table, where we can find a chronological list of web addresses involved in a file download, revealing the full chain of redirects and the final destination where the file was actually downloaded.

In our case, we have four entries for ID 5 , meaning that after clicking the link, there were three redirects before reaching the final URL:

[https://drive.usercontent.google.com/download?id=1z-SGnYJCPE0HA\\_Faz6N7mD5qf0E-A76H&export=download](https://drive.usercontent.google.com/download?id=1z-SGnYJCPE0HA_Faz6N7mD5qf0E-A76H&export=download) .

DB Browser for SQLite - /home/achille/Triage/C/Users/alonzo.spire/AppData/Local/Microsoft/Edge/User Data/Default/History

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project Attach Database Close Database

Browse Data Database Structure Edit Pragas Execute SQL

Table: downloads\_url\_chains Filter in any column

|   | id   | chain_index | url                                                                                                                                   |
|---|------|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
|   | F... | Filter      | Filter                                                                                                                                |
| 1 | 1    | 0           | https://download.sysinternals.com/files/PSTools.zip                                                                                   |
| 2 | 2    | 0           | https://uc5a544c88c0224e03ac1206c6f7.dl.dropboxusercontent.com/cd/0/get/CPH-3-R9QXZOmuk-aTzFfA5aEfMZIJ01DTQCPTSgu1cPjciRmjaQDjwRt-... |
| 3 | 3    | 0           | blob:https://github.com/73a5af0a-00b5-4915-969f-53dc3a0fcfd4                                                                          |
| 4 | 4    | 0           | blob:https://github.com/a2396f10-69ae-42ac-90db-89fbd0ca373                                                                           |
| 5 | 5    | 0           | https://l.facebook.com/l.php?u=https%3A%2F%2Fdrive.usercontent.google.com%2Fu%2F%2Fuc%3Fid%3D1z-SGnYJCPE0HA_Faz6N7mD5qf0E-...         |
| 6 | 5    | 1           | https://drive.usercontent.google.com/u/2/uc?id=1z-SGnYJCPE0HA_Faz6N7mD5qf0E-A76H&export=download                                      |
| 7 | 5    | 2           | https://drive.usercontent.google.com/uc?id=1z-SGnYJCPE0HA_Faz6N7mD5qf0E-A76H&export=download                                          |
| 8 | 5    | 3           | https://drive.usercontent.google.com/download?id=1z-SGnYJCPE0HA_Faz6N7mD5qf0E-A76H&export=download                                    |

**Answer:** [https://drive.usercontent.google.com/download?id=1z-SGnYJCPE0HA\\_Faz6N7mD5qf0E-A76H&export=download](https://drive.usercontent.google.com/download?id=1z-SGnYJCPE0HA_Faz6N7mD5qf0E-A76H&export=download)

5. Alonzo then proceeded to install the newly download app, thinking that it is a legit AI tool. What is the true product version which was installed?

Since we know that Alonzo downloaded an archive related to Gemini AI, we will first try to search for the keyword `Gemini` in the Windows logs.

We use `chainsaw`, a very powerful tool to search and hunt on Windows logs:

```
chainsaw --no-banner search -e Gemini *
```

```

---
Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: MsiInstaller
    EventID_attributes:
      Qualifiers: 0
    EventID: 1042
    Version: 0
    Level: 4
    Task: 0
    Opcode: 0
    Keywords: '0x8000000000000000'
    TimeCreated_attributes:
      SystemTime: 2024-03-19T04:31:20.083769Z
    EventRecordID: 4779
    Correlation: null
    Execution_attributes:
      ProcessID: 0
      ThreadID: 0
    Channel: Application
    Computer: Forela-Wkstn001.forela.local
    Security_attributes:
      UserID: S-1-5-18
  EventData:
    Data:
      - C:\Users\alonzo.spire\Downloads\AI.Gemini Ultra For PC V1.0.1\Google AI Gemini Ultra For PC V1.0.1.msi
      - '9888'
      - (NULL)
      - (NULL)
      - (NULL)
      - (NULL)
      - ''
    Binary: null

[+] Found 2 hits
[achille@parrot] - [~/Triage/C/Windows/System32/winevt/logs]
$

```

There are two events, one of which is shown above:

- Both events come from the `MsiInstaller` provider.
- The two events, one with ID **1040** (Installer Started) and the other with ID **1042** (Installer Exited), are related to the installation of an installer named `Google AI Gemini Ultra For PC V1.0.1.msi`.
- The installer starts at `2024-03-19 04:31:02` and exited at `2024-03-19 04:31:20`.

We now try to get more information about this installer by filtering entries with provider equal to `MsiInstaller` and timestamp equal to or newer than `2024-03-19T04:31:02`.

We build the `chainsaw` query with the following options:

- `--timestamp` : Specifies which field contains the timestamp of the events.
- `-t` : The Tau expression that will filter events based on the specified provider



- `--from` : The timestamp we want to start from.

We need to use the same format for timestamps as shown in the log entry, so the `T` between date and time is required.

Finally, we add a `grep` command to display only the IDs of the events found and their timestamps:

```
chainsaw --no-banner search --timestamp
'Event.System.TimeCreated_attributes.SystemTime' -t
'Event.System.Provider_attributes.Name: MsiInstaller' --from '2024-
03-19T04:31:02' Application.evtx | grep -e EventID: -e SystemTime
```

```
[achille@parrot]~/Triage/C/Windows/System32/winevt/logs]
- $chainsaw --no-banner search --timestamp 'Event.System.TimeCreated_attributes.SystemTime' -t 'Event.System.Provider_attributes.Name: MsiInstaller'
--from '2024-03-19T04:31:02' Application.evtx | grep -e EventID: -e SystemTime
[+] Loading forensic artefacts from: Application.evtx
[+] Loaded 1 forensic files (5.3 MB)
[+] Searching forensic artefacts...
EventID: 1040
SystemTime: 2024-03-19T04:31:02.898151Z
EventID: 1042
SystemTime: 2024-03-19T04:31:20.083769Z
EventID: 11707
SystemTime: 2024-03-19T04:31:33.977693Z
EventID: 1033
SystemTime: 2024-03-19T04:31:33.977693Z
[+] Found 4 hits
[achille@parrot]~/Triage/C/Windows/System32/winevt/logs]
- $
```

There are a total of 4 events:

- **1040** and **1042** that we saw earlier
- **11707** (Product: <product name> -- Installation operation completed successfully.): This event is logged after a product is successfully installed.
- **1033** (Windows Installer installed the product.): This event contains information about the installed product, such as its name and version.

Now we can finally get the actual product version by printing the event from the **MsiInstaller** provider with ID **1033**:

```
chainsaw --no-banner search --timestamp
'Event.System.TimeCreated_attributes.SystemTime' -t
'Event.System.EventID: =1033' --from '2024-03-19T04:31:02'
Application.evtx
```

```

[achille@parrot]~/Triage/C/Windows/System32/winevt/logs]
$chainsaw --no-banner search --timestamp 'Event.System.TimeCreated_attributes.SystemTime' -t 'Event.System.EventID: =1033' --from '2024-03-19T04:31:02' Application.evtx
[*] Loading forensic artefacts from: Application.evtx
[*] Loaded 1 forensic files (5.3 MB)
[*] Searching forensic artefacts...
---
Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: MsiInstaller
    EventID_attributes:
      Qualifiers: 0
      EventID: 1033
      Version: 0
      Level: 4
      Task: 0
      Opcode: 0
      Keywords: '0x8000000000000000'
    TimeCreated_attributes:
      SystemTime: 2024-03-19T04:31:33.977693Z
      EventRecordID: 4783
      Correlation: null
    Execution_attributes:
      ProcessID: 0
      ThreadID: 0
      Channel: Application
      Computer: Forela-Wkstn001.forela.local
    Security_attributes:
      UserID: S-1-5-21-3239415629-1862073780-2394361899-1104
  EventData:
    Data:
      - Install
      - 1033
      - 1033
      - 0
      - Google
      - (NULL)
      - ''
    Binary: 7B41424332434530312D373841352D343535342D413332412D3434303241344538334242337D303030613566306465336336363234626336336331316339363863653362316464313430303030393034

[*] Found 1 hits
[achille@parrot]~/Triage/C/Windows/System32/winevt/logs]
$

```

Answer: 3.32.3

## 6. When was the malicious product/package successfully installed on the system?

To answer this question we will print the event with ID **11707** instead:

```

chainsaw --no-banner search --timestamp
'Event.System.TimeCreated_attributes.SystemTime' -t
'Event.System.EventID: =11707' --from '2024-03-19T04:31:02'
Application.evtx

```

```

[achille@parrot]~/Triage/C/Windows/System32/winevt/logs
$chainsaw --no-banner search --timestamp 'Event.System.TimeCreated_attributes.SystemTime' -t 'Event.System.EventID: =11707' --from '2024-03-19T04:31:02' Application.evtx
[*] Loading forensic artefacts from: Application.evtx
[*] Loaded 1 forensic files (5.3 MB)
[*] Searching forensic artefacts...
---
Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: MsInstaller
    EventID_attributes:
      Qualifiers: 0
      EventID: 11707
      Version: 0
      Level: 4
      Task: 0
      Opcode: 0
      Keywords: '0x8000000000000000'
    TimeCreated_attributes:
      SystemTime: 2024-03-19T04:31:33.977693Z
    EventRecordID: 4782
    Correlation: null
    Execution_attributes:
      ProcessID: 0
      ThreadID: 0
      Channel: Application
      Computer: Forela-Wkstm001.forela.local
    Security_attributes:
      UserID: 5-1-5-21-3239415629-1862073780-2394361899-1104
EventData:
  Data:
    'Product: Install -- Installation completed successfully.'
    - (NULL)
    - (NULL)
    - (NULL)
    - (NULL)
    - (NULL)
    - ''
  Binary: 7B41424332434530312D373841352D343535342D413332412D3434303241344538334242337D
[*] Found 1 hits
[achille@parrot]~/Triage/C/Windows/System32/winevt/logs

```

**Answer:** 2024-03-19 04:31:33

## 7. The malware used a legitimate location to stage its file on the endpoint. Can you find out the Directory path of this location?

To answer this question, we will focus our attention on the \$MFT artifact. After converting it to a CSV file, we open it with **Timeline Explorer**.

Then we apply a time filter to select only events with FileName creation timestamp ( Created0x30 ) newer than the first installation event of the MSI file.

After sorting the entries in ascending order, we can see right away that there are a bunch of files that are located under the C:\Program Files (x86)\Google\Install folder.

Timeline Explorer v2.1.0  
File Tools Tabs View Help  
inf.csv

Drag a column header here to group by that column

Enter text to search... Find

| Parent Path                                                                                     | File Name                                      | Extension | Is Directory                        | Created0x30           | Has Ads                             | Is A |
|-------------------------------------------------------------------------------------------------|------------------------------------------------|-----------|-------------------------------------|-----------------------|-------------------------------------|------|
| .\Program Files (x86)\Google\Install\nmmhkkgccagldgiimedpic                                     | background.js                                  | .js       | <input type="checkbox"/>            | > 2024-03-19 04:31:20 | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\New Folder #%d2                                            | account_manager (22).xls                       | .xls      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install                                                            | System.Deployment.dll                          | .dll      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install                                                            | Microsoft.VisualBasic.dll                      | .dll      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\nmmhkkgccagldgiimedpic                                     | favicon.png                                    | .png      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install                                                            | System.Web.DynamicData.Design.dll              | .dll      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\nmmhkkgccagldgiimedpic                                     | manifest.json                                  | .json     | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\nmmhkkgccagldgiimedpic                                     | ru.ps1                                         | .ps1      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\New Folder #%d2                                            | account_manager (21).xls                       | .xls      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install                                                            | install.cmd                                    | .cmd      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\New Folder #%d2                                            | list_page (3).xlsx                             | .xlsx     | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\New Folder #%d2                                            | account_manager (20).xls                       | .xls      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install                                                            | logo.ico                                       | .ico      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Program Files (x86)\Google\Install\nmmhkkgccagldgiimedpic                                     | content.js                                     | .js       | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Windows\Installer                                                                             | 120fe8.msi                                     | .msi      | <input type="checkbox"/>            | 2024-03-19 04:31:20   | <input type="checkbox"/>            |      |
| .\Windows\WinSxS\wow64_microsoft-windows-r...xuddmdriver-wow64-c_31bf3856ad364e35_10.0.19041... | openc1.dll                                     | .dll      | <input type="checkbox"/>            | 2024-03-19 04:32:49   | <input checked="" type="checkbox"/> |      |
| .\Windows\WinSxS\wow64_microsoft-windows-r...xuddmdriver-wow64-c_31bf3856ad364e35_10.0.19041... | openc1.dll:WofCompressedData                   |           | <input type="checkbox"/>            | 2024-03-19 04:32:49   | <input type="checkbox"/>            |      |
| .\Windows\WinSxS\amd64_microsoft-windows-s...defaultassociations_31bf3856ad364e35_10.0.19041... | OEMDefaultAssociations.xml                     | .xml      | <input type="checkbox"/>            | 2024-03-19 04:33:02   | <input checked="" type="checkbox"/> |      |
| .\Windows\WinSxS\amd64_microsoft-windows-s...defaultassociations_31bf3856ad364e35_10.0.19041... | OEMDefaultAssociations.xml:WofCompressedData   |           | <input type="checkbox"/>            | 2024-03-19 04:33:02   | <input type="checkbox"/>            |      |
| .\Windows\WinSxS\amd64_microsoft-windows-s...defaultassociations_31bf3856ad364e35_10.0.19041... | openc1.dll:WofCompressedData                   | .dll      | <input type="checkbox"/>            | 2024-03-19 04:33:02   | <input checked="" type="checkbox"/> |      |
| .\Windows\WinSxS\amd64_microsoft-windows-s...defaultassociations_31bf3856ad364e35_10.0.19041... | OEMDefaultAssociations.dll:WofCompressedData   |           | <input type="checkbox"/>            | 2024-03-19 04:33:02   | <input type="checkbox"/>            |      |
| .\ProgramData\Microsoft\Windows\DeviceMetadataCache                                             | dmrc.idx                                       | .idx      | <input type="checkbox"/>            | 2024-03-19 04:38:11   | <input type="checkbox"/>            |      |
| .\Users\alanzo.spire\AppData\Local\Temp\vmware-alonzo.spire\VMwareDnD\ac06a20c                  | KAPE.zip                                       | .zip      | <input type="checkbox"/>            | 2024-03-19 04:40:04   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin                                                           | S-1-5-21-3239415629-1862073780-2394361899-1... |           | <input checked="" type="checkbox"/> | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103            | \$1009F03                                      |           | <input type="checkbox"/>            | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103            | \$12NDIFG.tmp                                  | .tmp      | <input type="checkbox"/>            | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103            | \$13VQ050.tmp                                  | .tmp      | <input type="checkbox"/>            | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103            | \$152MPJ0.exe                                  | .exe      | <input type="checkbox"/>            | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |
| .\Windows\Temp\Triage\CL\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103            | \$15MGYSD                                      |           | <input type="checkbox"/>            | 2024-03-19 04:43:18   | <input type="checkbox"/>            |      |

X Created0x30 > 2024-03-19 04:31:20 Edit Filter

C:\Users\achille\Desktop\inf.csv Total lines 603,019 Visible lines 342 Open files: 1 Search options

Answer: C:\Program Files (x86)\Google

## 8. The malware executed a command from a file. What is name of this file?

From the previous analysis we saw that there are some suspicious files in the identified folder, such as `ru.ps1`, `install.cmd`, `background.js` and `content.js`.

Since this question asks from which file a command was executed, the most likely is `install.cmd`.

Answer: `install.cmd`

## 9. What are the contents of the file from question 8? Remove whitespace to avoid format issues.

We can confirm our previous assumption by looking at the contents of `install.cmd` using **MFTECmd**.

From the **Timeline Explorer** we noted the Entry Number (51471) and the Sequence Number (4) for `install.cmd`, then we execute:

```
.\MFTECmd.exe -f '..\MFT' --de 51471-4
```

[illegible]

```
**Answer:** <span style="color: #9FEF00;">`@echooffpowershell-
ExecutionPolicyBypass-
File"%~dp0nmhkkeqccagldqiimedpic/ru.ps1"`</span>
```

10. What was the command executed from this file according to the logs?

Because this is a **PowerShell** command, we expect to see the execution of this command in the **Windows PowerShell** event logs.

Again, we use `chainsaw` to find this entry in the EVTX file:

```
chainsaw --no-banner search -e 'ru.ps1' logs/Windows\
PowerShell.evtx | tail -n 30
```

We need to remove the escape character ( \ ) from the path and then we get:

```
powershell -ExecutionPolicy Bypass -File C:\Program Files  
(x86)\Google\Install\nmmhkkeagccagdldqiimedpic/ru.ps1.
```

```

[+] Loading forensic artefacts from: logs/Windows PowerShell.evtx | tail -n 30
[+] Loaded 1 forensic files (1.1 MB)
[+] Searching forensic artefacts...
[+] Found 8 hits
xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
System:
  Provider_attributes:
    Name: PowerShell
  EventID_attributes:
    Qualifiers: 0
    EventID: 483
    Version: 0
    Level: 4
    Task: 4
    Opcode: 0
    Keywords: '@x8000000000000000'
  TimeCreated_attributes:
    SystemTime: 2024-03-19T04:31:22.552551Z
    EventRecordID: 261
    Correlation: null
  Execution_attributes:
    ProcessID: 0
    ThreadID: 0
    Channel: Windows PowerShell
    Computer: Forela-Wkstn001:forela.local
    Security: null
  EventData:
    Data:
      - Stopped
      - Available
      - {
        "id": 1,
        "name": "PowerShell - ExecutionPolicy Bypass - File C:\\Program Files (x86)\\Google\\Install\\nmmhkkegccagdldgiimedpic\\ru.ps1",
        "command": "powershell -ExecutionPolicy Bypass -File C:\\Program Files (x86)\\Google\\Install\\nmmhkkegccagdldgiimedpic\\ru.ps1",
        "commandType": "Script",
        "commandPath": "C:\\Program Files (x86)\\Google\\Install\\nmmhkkegccagdldgiimedpic\\ru.ps1",
        "binary": null
      }
    }
[+] achille@parrot: ~/Triage/C/Windows/System32/winevt

```

**Answer:** powershell -ExecutionPolicy Bypass -File C:\Program Files (x86)\Google\Install\nmmhkkegccagdldgiimedpic/ru.ps1

11. Under malware staging Directory, a js file resides which is very small in size. What is the hex offset for this file on the filesystem?

We have already identified two JS files: background.js and content.js , which are 17208 and 258 bytes respectively.

So again, from the **Timeline Explorer**, we note the Entry and Sequence numbers for content.js and then pass them to **MFTCmd**:

```
.\MFTCmd.exe -f '..\MFT' --de 64067-4
```

The output will provide us with the Offset ( 3E90C00 ) and also the content of the file.

```
Windows PowerShell
PS C:\Users\achille\Desktop\tools> .\MFTECmd.exe -f '..\SMFT' --de 64067-4
MFTECmd version 1.3.0.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/MFTECmd

Command line: -f ..\SMFT --de 64067-4

Warning: Administrator privileges not found!

File type: Mft

Processed ..\SMFT in 30.2950 seconds

..\SMFT: FILE records found: 489,505 (Free records: 1,169) File size: 479.2MB

Dumping details for file record with key 0000FA43-00000004
Entry-seq #: 0xFA43-0x4, Offset: 0x3E90C00, Flags: 1, Log seq #: 0x1522B4F45, Base Record entry-seq: 0x0-0x0
Reference count: 0x1, FixUp Data Expected: 06-00, FixUp Data Actual: 74-53 | 00-00 (FixUp OK: True)

**** STANDARD INFO ****
Attribute #: 0x0, Size: 0x60, Content size: 0x48, Name size: 0x0, ContentOffset 0x18. Resident: True
Flags: Archive, Max Version: 0x0, Flags 2: None, Class Id: 0x0, Owner Id: 0x0, Security Id: 0x226, Quota charged: 0x0, Update sequence #: 0x53CE390

Created On: 2024-01-22 13:59:20.0000000
Modified On: 2024-01-22 13:59:20.0000000
Record Modified On: 2024-03-19 04:31:20.0371820
Last Accessed On: 2024-03-19 04:31:24.0130269

**** FILE NAME ****
Attribute #: 0x2, Size: 0x70, Content size: 0x56, Name size: 0x0, ContentOffset 0x18. Resident: True
File name: content.js
Flags: Archive, Name Type: DosWindows, Reparse Value: 0x0, Physical Size: 0x0, Logical Size: 0x0
Parent Entry-seq #: 0xC897-0x4

Created On: 2024-03-19 04:31:20.0371820
Modified On: 2024-03-19 04:31:20.0371820
Record Modified On: 2024-03-19 04:31:20.0371820
Last Accessed On: 2024-03-19 04:31:20.0371820

**** DATA ****
Attribute #: 0x1, Size: 0x120, Content size: 0x102, Name size: 0x0, ContentOffset 0x18. Resident: True

Resident Data
Data: 76-61-72-20-69-73-43-6F-6E-74-65-6E-74-53-63-72-69-70-74-45-78-65-63-75-74-65-64-20-3D-20-6C-6F-63-61-6C-53-74-6F-72-61-67-65-2E-67-65-74-49-
74-65-6D-28-27-63-6F-6E-74-65-6E-74-53-63-72-69-70-74-45-78-65-63-75-74-65-64-27-29-3B-0D-0A-69-66-20-28-21-69-73-43-6F-6E-74-65-6E-74-53-63-72-69-70-
74-45-78-65-63-75-74-65-64-29-20-7B-0D-0A-63-68-72-6F-6D-65-2E-72-75-6E-74-69-6D-65-2E-73-65-6E-64-4D-65-73-73-61-67-65-28-7B-20-61-63-74-69-6F-6E-3
A-20-27-65-78-65-63-75-74-65-46-75-6E-63-74-69-6F-6E-27-20-7D-2C-20-66-75-6E-63-74-69-6F-6E-20-28-72-65-73-70-6F-6E-73-65-29-20-7B-0D-0A-20-20-6C-6F-
63-61-6C-53-74-6F-72-61-67-65-2E-73-65-74-49-74-65-6D-28-27-63-6F-6E-74-65-6E-74-53-63-72-69-70-74-45-78-65-63-75-74-65-64-27-2C-20-74-72-75-65-29-3B-
0D-0A-7D-29-3B-0D-0A-7D-0D-0A-0D-0A

ASCII: var isContentScriptExecuted = localStorage.getItem('contentScriptExecuted');
if (!isContentScriptExecuted) {
chrome.runtime.sendMessage({ action: 'executeFunction' }, function (response) {
localStorage.setItem('contentScriptExecuted', true);
});
}
```

Answer: 3E90C00

12. Recover the contents of this js file so we can forward this to our RE/MA team for further analysis and understanding of this infection chain. To sanitize the payload, remove whitespaces.

In the previous question we also found the contents of the file:

```
var isContentScriptExecuted =
localStorage.getItem('contentScriptExecuted');
if (!isContentScriptExecuted) {
chrome.runtime.sendMessage({ action: 'executeFunction' }, function
(response) {
    localStorage.setItem('contentScriptExecuted', true);
});
}
```

By removing the whitespaces we get the correct answer.

Answer:

```
var isContentScriptExecuted=localStorage.getItem('contentScriptExecuted');if(!isContentScriptExecuted){chrome.runtime.sendMessage({action:'executeFunction'},function(response){localStorage.setItem('contentScriptExecuted',true);});}
```

13. Upon seeing no AI Assistant app being run, alonzo tried searching it from file explorer. What keywords did he use to search?

This is not a common DFIR technique, but there is a great [article](#) that explains how to retrieve information about searched strings in Windows.

So we fire up **Registry Explorer** and feed it with Alonzo's NTUSER.DAT .

Then browsing to

\Software\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery we can see the keyword that Alonzo searched: Google Ai Gemini .

The screenshot shows the Registry Explorer v2.1.0 interface. The left pane displays the tree structure of the registry, with the path \Software\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery expanded. The right pane shows the 'Values' tab for 'WordWheelQuery'. A table lists the registry values, with one entry highlighted: 'Google Ai Gemini tool' with a value of '0' and a last write timestamp of '2024-03-19 04:32:11'. Below the table, a hex view of the data is shown, displaying '00 00 00 00 FF FF FF FF'.

| Search Term           | Mru Position | Key Name       | Last Write Timestamp |
|-----------------------|--------------|----------------|----------------------|
| Google Ai Gemini tool | 0            | WordWheelQuery | 2024-03-19 04:32:11  |

Answer: Google Ai Gemini tool



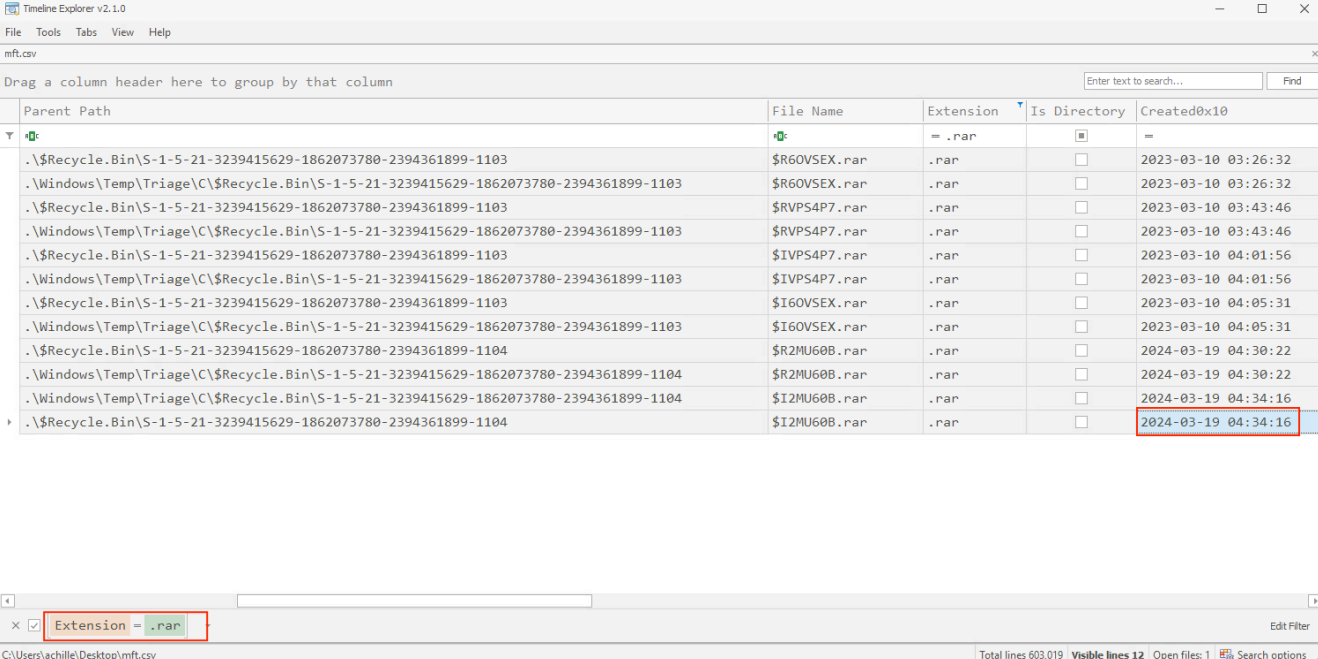
#### 14. When did alonzo searched it?

The Last Write Timestamp column in the previous screenshot gives us the answer:  
2024-03-19 04:32:11 .

**Answer:** 2024-03-19 04:32:11

#### 15. After alonzo could not find any AI tool on the system, he became suspicious, contacted the security team and deleted the downloaded file. When was the file deleted by alonzo?

We switch back to our **Timeline Explorer** and, since Alonzo downloaded a RAR archive, this time we apply a filter to select only files with a .rar extension. Then we sort the timestamps in ascending order and the only one older than the download time is: 2024-03-19 04:34:16 .



| Parent Path                                                                          | File Name     | Extension | Is Directory             | Created0x10         |
|--------------------------------------------------------------------------------------|---------------|-----------|--------------------------|---------------------|
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103                       | \$R60VSEX.rar | .rar      | <input type="checkbox"/> | 2023-03-10 03:26:32 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103 | \$R60VSEX.rar | .rar      | <input type="checkbox"/> | 2023-03-10 03:26:32 |
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103                       | \$RVPS4P7.rar | .rar      | <input type="checkbox"/> | 2023-03-10 03:43:46 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103 | \$RVPS4P7.rar | .rar      | <input type="checkbox"/> | 2023-03-10 03:43:46 |
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103                       | \$IVPS4P7.rar | .rar      | <input type="checkbox"/> | 2023-03-10 04:01:56 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103 | \$IVPS4P7.rar | .rar      | <input type="checkbox"/> | 2023-03-10 04:01:56 |
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103                       | \$I60VSEX.rar | .rar      | <input type="checkbox"/> | 2023-03-10 04:05:31 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1103 | \$I60VSEX.rar | .rar      | <input type="checkbox"/> | 2023-03-10 04:05:31 |
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1104                       | \$R2MU60B.rar | .rar      | <input type="checkbox"/> | 2024-03-19 04:30:22 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1104 | \$R2MU60B.rar | .rar      | <input type="checkbox"/> | 2024-03-19 04:30:22 |
| .\Windows\Temp\Triage\C\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1104 | \$I2MU60B.rar | .rar      | <input type="checkbox"/> | 2024-03-19 04:34:16 |
| .\\$Recycle.Bin\S-1-5-21-3239415629-1862073780-2394361899-1104                       | \$I2MU60B.rar | .rar      | <input type="checkbox"/> | 2024-03-19 04:34:16 |

Filter: Extension = .rar

**Answer:** 2024-03-19 04:34:16

#### 16. Looking back at the starting point of this infection, please find the MD5 hash of the malicious installer.

To answer this last question, we try to search for the MSI file name on the web. The first result is a page from **AnyRun** (a well-known online sandbox) that analyzes this installer, so we can get the MS5 hash of the file from here:

BF17D7F8DAC7DF58B37582CEC39E609D .

ANY.RUN

INTERACTIVE MALWARE ANALYSIS

General

Behavior

MalConf

Static information

Video

Screenshots

System events

Network

General Info

☒ Add for printing

File name:

Google AI Gemini Ultra For PC V1.0.1.msi

Full analysis:

<https://app.any.run/tasks/0ef391d1-20c0-4279-adb6-e89afe28d37f>

Verdict:

Malicious activity

Analysis date:

February 11, 2024 at 23:33:59

OS:

Windows 10 Professional (build: 19044, 64 bit)

Indicators:

MIME:

application/x-msi

File info:

Composite Document File V2 Document, Little Endian, Os: Windows, Version 10.0, MSI Installer, Create Time/Date: Mon Jun 21 07:00:00 1999, Name of Creating Application: Windows Installer, Security: 1, Code page: 1252, Template: Intel;1033, Number of Pages: 200, Revision Number: {C6C2D440-EA93-4E35-B8C6-83DD50C1613C}, Title: Install, Author: Install, Comments: Bringing the benefits of AI to everyone, Number of Words: 2, Last Saved Time/Date: Mon Jan 22 11:59:18 2024, Last Printed: Mon Jan 22 11:59:18 2024

MD5:

BF17D7F8DAC7DF58B37582CEC39E609D

SHA1:

0C55B3C75E5759EFC6DB20B6DB4FAD790C8CD4E7

SHA256:

BB7C3B78F2784A7AC3C090331326279476C748087188AEB69F431BBD70AC6407

SSDEEP:

24576:C1ipiRvE4wbF60m3o0EZ/vTk1oQ53eYc:C1ipiRM4wbF60m3o0EZ/vTk1oQ53lc

ANY.RUN is an interactive service which provides full access to the guest system. Information in this report could be distorted by user actions and is provided for user acknowledgement as it is. ANY.RUN does not guarantee maliciousness or safety of the content.

Answer: BF17D7F8DAC7DF58B37582CEC39E609D